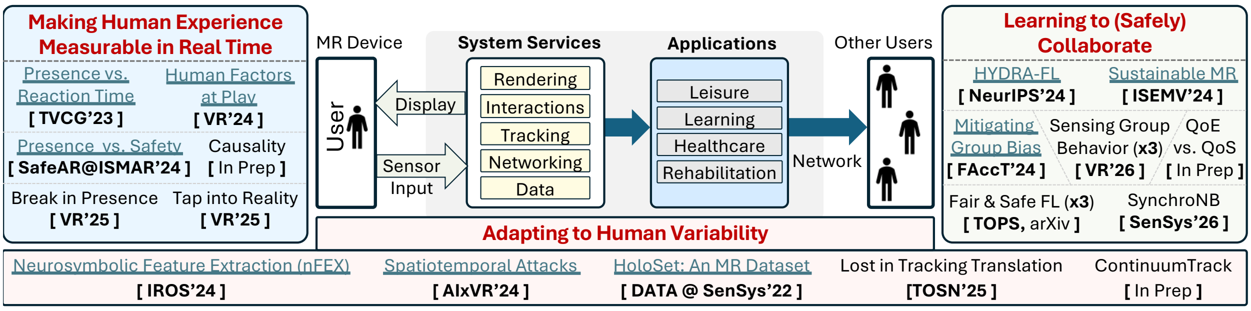
Research Statement

Yasra Chandio
yasrachandio.github.io

I am a Human-Centered Computing researcher who develops systems that capture and translate human signals into data representations, which artificial intelligence (AI) models require to learn from and adapt to human variability and experience. My work spans `Human-Computer Interaction (HCI)`, `Systems`, and `Machine Learning (ML)`, and focuses on integrating human factors into the pipeline of sensing, modeling, and interaction. I develop methods to interpret subjective human experiences and variability, and collaboration, turning them into actionable inputs for adaptive AI.



Mixed Reality (MR) serves as my application domain because it exposes the challenges AI systems face when interacting with people, but the methods generalize well beyond. MR combines physical and virtual worlds by sensing the user's environment and interactions, using position tracking to align virtual elements with real objects. The system processes this tracking data so virtual content responds in real time, producing experiences that feel both believable and safe, creating a sense of immersion. This creates the experiential core of MR, which is presence, the feeling of "being there" in the virtual environment. Presence is both a technical milestone and a human outcome, and its effect depends on the downstream application, whether therapy motivates recovery, training builds confidence, or collaboration sustains trust. These examples illustrate a broader AI challenge, where systems must adapt to uncertainty, variability, and social context.

Across domains, the demands differ. Games prioritize responsiveness and visual quality, surgical tools require precise tracking for patient safety, and collaborative platforms must balance resources, privacy, security, and fairness so no participant undermines the collective experience. These challenges define the focus of my research, advancing along three thrusts connecting technical design to human impact.

(1) Creating `experiential metrics` that capture signals of human state, such as using reaction time to reflect presence and cognitive load, and applying causal inference to test which signals drive changes in perception and performance [1–7], enabling systems to adapt when users become distracted, fatigued, or overloaded.

(2) Building `adaptive tracking methods` that remain reliable across the natural variability of human movement [8, 9], supporting applications such as rehabilitation, where irregular gestures and fatigue are part of recovery.

(3) Developing collaborative sensing systems that passively capture multimodal data to `infer group dynamics` [10–12], enabling real-time detection of collaboration patterns and `supporting safe and fair collaboration` [13–17] with privacy preserving federated learning.

1 Making Human Experience Measurable in Real Time *(VR×3, TVCG, SafeAR@ISMAR)*.

Experiential Thrust asks how interactive systems can recognize what people feel and adapt to keep them engaged, safe, and included. I develop experiential metrics that connect human perception, cognitive state, and real-time interaction, because improving adaptive systems requires measuring human experience accurately. I identified reaction time (the time it takes for a user to respond to a cue) as a key measure of presence (a subjective feeling that reflects the user's immersive experience). Reaction time is readily available and reveals how visual, auditory, and task demands shape the user experience, making it possible to adapt environments in real-time for patients in rehabilitation, students in classrooms, or teams collaborating remotely.

I showed that reaction time is influenced by sensory design and human cognition. By investigating the *link between the human's cognitive state and their performance* [1], I found that factors like familiarity and task conditioning shape how people respond. This demonstrated that reaction time reflects both the fidelity of the scene and the user's cognitive state. By *linking human perception (subjective experiential element) to human actions (systemic performance indicator)* [2], my work gave designers a tool to see how their choices directly affect what users feel and do, allowing training environments to build confidence and therapy tools to adjust difficulty without overwhelming patients. Building on this, I evaluated how interaction design impacts presence, using reaction time to *assess presence (experiential) and fine-tune interactions (systemic)* to improve the overall experience and design [3]. Intuitive interactions that mimic the real world reduced cognitive load and enhanced presence, while abstract ones increased load, slowed responses, and diminished experience. These results highlight the importance of designing *interactions that are both perceptually intuitive and cognitively manageable* for users [5], enabling access across diverse cognitive and physical needs [6].

Finally, I studied how everyday interruptions disrupt immersion. Congruent distractions, such as a familiar sound, were easier to rationalize, while incongruent ones caused cognitive overload, longer reaction times, and loss of presence. I quantified this by modeling the *mediation* between distraction, cognitive load, reaction time, and presence [4], showing how disruptive events threaten safety and delay recovery. Correlation alone is not enough to explain these effects, so I applied the causal inference to test which *experiential cues genuinely drive perception and performance* [7]. This causal framing advances the AI principle of moving from correlation to intervention and counterfactual reasoning, identifies which experiential cues matter most for adaptation (`when to act`), and provides AI systems with principled inputs to respond proactively (`how to act`) in sensitive domains such as surgery, therapy, or collaborative work, where lapses in attention can have lasting consequences.

2 Adapting to Human Variability *(IROS, AIxVR, TOSN, DATA@SenSys, LBW@Sensys).*
Systemic Thrust contributes to core infrastructure for interactive AI systems by improving pose tracking accuracy, adaptability, and security. In these systems, users move fluidly between settings such as indoors and outdoors under varied lighting while interacting with both physical and virtual elements. Tracking failures in such environments are not just technical glitches; they add cognitive strain, disrupt interaction, and in safety-critical contexts, compromise physical safety.

To address this, I developed Neurosymbolic Feature Extraction (`nFEX`), an adaptive framework for *human-centric feature extraction for pose tracking* [8]. `nFEX` combines the clarity of symbolic reasoning with the adaptability of neural networks, enabling AI systems to respond to changes in lighting, textures, and user motion. To support it, I created two resources: symbolic rules that inform the design of *tracking solutions suited to unpredictable human interactions* [9], and the HoloSet dataset *capturing previously unexplored human interactions* [18]. With ∼100K multimodal samples, HoloSet validated nFEX's adaptability, showing up to 90% error reduction in complex scenarios, directly improving safety and usability in rehabilitation, surgery, and collaborative learning.

Even with adaptability, interactive AI systems remain vulnerable to stealthy manipulations. I uncovered *a novel attack surface* that exploits the multimodal and spatiotemporal nature of tracking systems [19, 20], demonstrating how subtle manipulations can quietly degrade accuracy and threaten safety. To complement `nFEX`'s real-time adaptability, I introduced *ContinuumTrack* [21], a system that *anticipates and resolves tracking failures before they occur*, combining profiling, prediction, and correction to sustain long-term reliability. Together, these advances support AI systems that are immersive, trustworthy, and safe in interactive and safety-critical domains.

3 Learning to (Safely) Collaborate *(NeurIPS, FAccT, Sensys, eEnegry, TOPS, ISEMV).*
User Collaboration Thrust addresses fairness, privacy, security, and resource efficiency in shared interactive systems to promote equitable interactions between users. My contributions include

developing multimodal sensing techniques to *sense and interpret individual behaviors in a group* to infer group dynamics [10], such as identifying whether users are collaborating, competing, or working independently. I built a prototype that passively captured user behavior and applied sociometric analysis using unsupervised learning to *infer group activity in real time* [11], determining whether individual perception of collaboration matches the actual group experience [12]. These methods help interactive systems adapt to collaboration patterns and prevent individuals from being excluded.

However, continuous sensing and readily available data also raise security and privacy concerns, since systems may capture sensitive conversations or personal habits. Disproportionate data collection from users with particular speech patterns or gestures can introduce biases, leading to misinterpretation and unfair treatment. Malicious users manipulating shared resources could inject biased data, expose private interactions, or cause security breaches, undermining collaboration. To address these challenges, I applied privacy-preserving learning methods. Through Federated Learning, my work supports *secure and fair data handling* across multiple users [**Khan2025FederatedLearningPitfalls**, 13–15, 22, 23], enabling fair model training without centralizing sensitive data. These contributions support applications in classrooms, therapy groups, and distributed collaboration platforms where individual support should strengthen rather than disrupt the shared experience.

Equitable collaboration also depends on performance and sustainability. I studied the *critical trade-offs between quality-of-service (QoS) and quality-of-experience (QoE)* [24] and advanced methods to minimize the carbon footprint and enhance *resource efficiency without sacrificing performance* [25–27]. These advances make collaborative AI systems both inclusive and environmentally responsible, supporting equitable participation while aligning with broader societal goals.

Future Vision

Current platforms assume neurotypical behavior, treat variability as error, and offer little transparency to users. My vision is to advance `Human-Centered AI`, where systems place human experience at the center of adaptation. Presence, cognitive load, movement variability, and collaboration dynamics become signals that guide AI, with participation as the outcome. The next phase of my work will extend `from controlled studies into real-world clinical and educational deployments` in therapy, rehabilitation, classrooms, and health, where adaptive AI can demonstrate tangible impact.

1 **Inclusive Sensing of Cognitive and Emotional States:** My work has shown that reaction time can capture presence and cognitive load in real time. This creates the foundation for AI systems that *use experiential and affective cues to guide adaptation*. Expanding beyond reaction time, I will develop richer multimodal sensing pipelines that incorporate gaze, voice, and facial expressions. These will model how people think and feel during interactions, creating systems that are perceptually, physically, and socially aware.

The **short-term goal** is to design systems that monitor explicit feedback (reaction time) alongside implicit cues (gaze shifts, speech prosody, facial expressions). These prototypes will extend my current methods with multimodal sensing and lightweight learning models to detect disengagement, overload, or stress as they happen. These systems will support applications such as classroom learning, where early detection of attention loss can prevent cascading disengagement, cognitive overload, or physical disorientation, all of which pose a critical risk.

The **long-term goal** is to develop *causal models for cognitive and emotional states* that learn from both neurotypical and neurodivergent populations. These models will identify which signals genuinely drive changes in performance and which are incidental. Paired with conversational generative AI, they will enable adaptive systems that explain interventions in real time, helping neurodivergent learners stay engaged while giving them control over their learning experience.

2 **Learning from Human Variability:** Current systems often treat variability in human behavior as error, but my work has shown that variability in movement and attention is a critical context for adaptation. Building on my contributions in adaptive tracking and experiential metrics, I will

design platforms that *treat human variability as a learning signal rather than noise*, especially in rehabilitation, where recovery depends on irregular and changing patterns. The question I aim to answer is how to account for individual variability in users' responses and tailor safety interventions to meet each user's needs.

The **short-term goal** is to develop proof-of-concept systems that integrate experiential signals such as reaction time, gaze, and irregular motion into adaptive feedback loops for rehabilitation applications. These prototypes will adjust exercises in response to fatigue or loss of focus, sustaining patient motivation and preventing discouragement.

The **long-term goal** is to scale this approach to *continual learning models of recovery trajectories*. I will create rehabilitation systems that integrate motor, cognitive, and emotional signals into comprehensive recovery models that evolve with the patient and generalize across diverse conditions. These systems will evolve with each patient's progress, personalizing therapy that adapts over weeks and months. By embracing variability, they will provide scalable rehabilitation tools that remain sensitive to individual recovery paths.

3 **Equitable and Symmetrical Collaboration:** Collaboration introduces challenges of fairness, privacy, and efficiency. My work on multimodal sensing and federated learning demonstrates how group interactions can be captured and adapted in real time. I will extend this foundation to design AI systems that support *fair, secure, and sustainable participation in groups*, with a focus on older adults who are at risk of exclusion in collaborative settings. For neurodivergent populations, these platforms can both interpret and explain the interactions of others in clearer ways and adapt to their own experiences in real time, so that group work remains accessible and inclusive.

The **short-term goal** is to build prototypes that apply unsupervised learning to multimodal group signals (speech balance, gaze overlap, proximity) to detect participation gaps, mitigate bias, and defend against malicious actors. These systems will provide immediate feedback that helps older adults stay engaged and allows neurodivergent users to participate without being overshadowed or excluded. To prevent any user from dominating or inducing malicious interactions in a group setting.

The **long-term goal** is to develop *fairness-aware federated models for collaboration* that preserve privacy while supporting personalization across diverse groups and platforms. I will use causal inference models to explain why inequities or subgroup biases arise, identify mechanisms of decline, and allow interventions that are principled and transparent. Over time, these platforms will scale to support healthy aging and inclusive group work by combining fairness with resource- and carbon-aware optimization, ensuring collaboration remains both equitable and sustainable.

My agenda combines theoretical contributions, system prototypes, and human subject studies. I will expand interdisciplinary collaborations, building on work with cognitive scientists and extending to neuroscience and clinical partners. A cross-cutting element will be conversational generative AI, making adaptation transparent by explaining changes, scaffolding tasks, and offering reassurance. My long-term vision is `MR as a human-centered AI medium that centers human experience` and creates opportunities for participation across diverse cognitive and physical needs.

Broader Impact

While my research advances MR systems, it has far-reaching societal implications. The behavioral and safety tools I develop in my experiential and systemic thrusts extend to accessibility and healthcare, aging, and align with NIH goals. My real-time monitoring methods can support neurodivergent individuals and those with autism through personalized feedback, track cognitive and physical changes in aging for early intervention, and monitor activity and diet to encourage healthier choices in diabetes management [1]. With my interdisciplinary background, I find both the NSF and NIH CAREER programs highly relevant to my research trajectory.

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